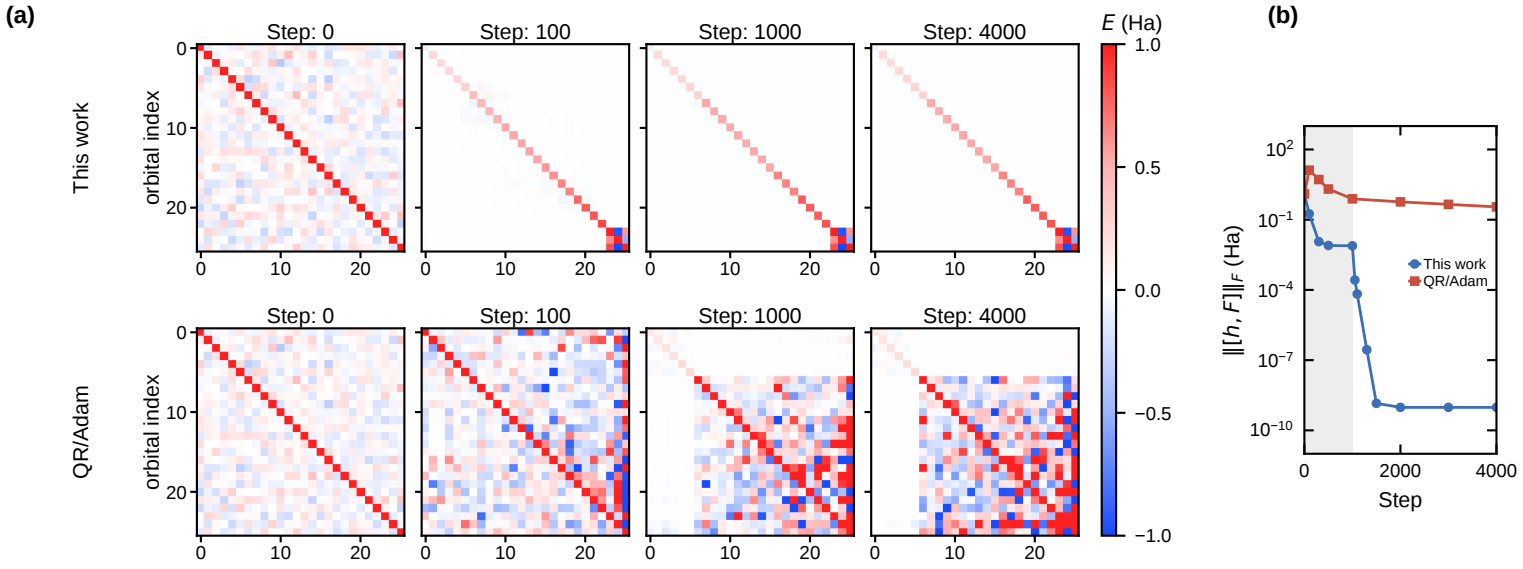
(a)
This work
QR/Adam
Step: 0
Step: 100
Step: 1000
Step: 4000
orbital index
0
10
20
E (Ha)
1.0
0.5
0.0
−0.5
−1.0
(b)
$\|[h, F]\|_F$ (Ha)
10^{2}
10^{-1}
10^{-4}
10^{-7}
10^{-10}
This work
QR/Adam
0
2000
4000
Step